Linear equations and inequalities

1 $x \geq -3$

2 $x < 6$

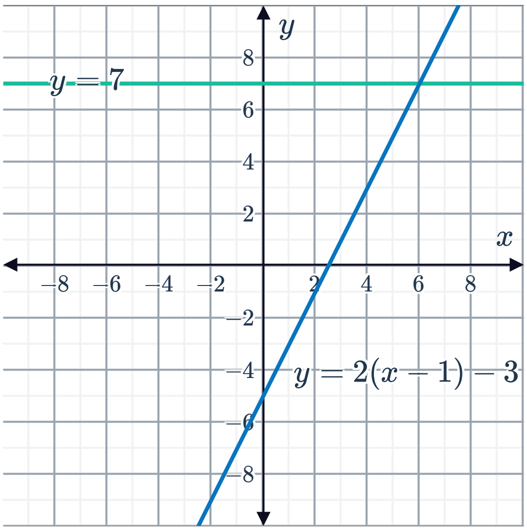
3 $\left(-\infty, -\frac{3}{5}\right]$

4

a $x = 6$

b $y = 7$ and $y = 2\left(x - 1\right) - 3$

c

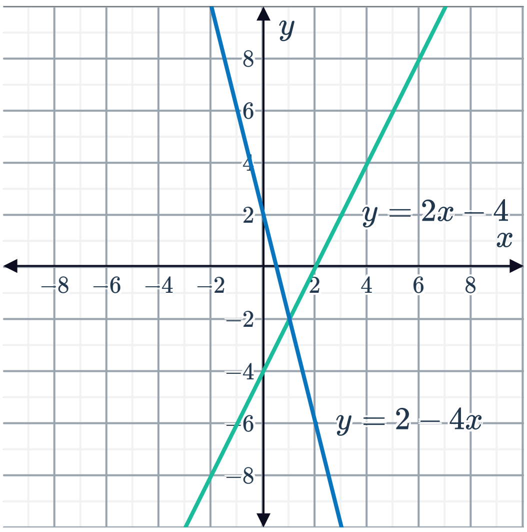


d $x = 6$

5 Infinite number of solutions

6

a



b $(1, -2)$

c $x > 1$

8 $\left(\frac{7}{2}, \infty\right)$

9

a $x > 3$

b $x \leq -4$

c $x \leq 27$

d $x < -4$

Quadratic equations and inequalities

10

a $x = 6, 12$

b $6 < x < 12$

c $x < 6, x > 12$

d $x = 9$

11 $-1 < x < 5$

12

a All real x, such that $x \neq 4$.

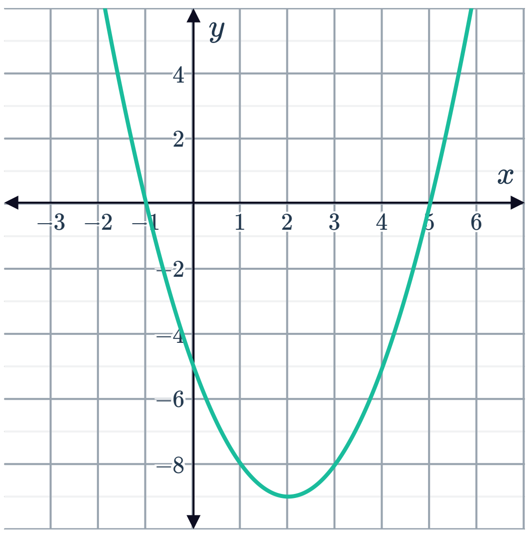
b $x = 4$

c $x = 4$

d $\triangle = 0$

13

a

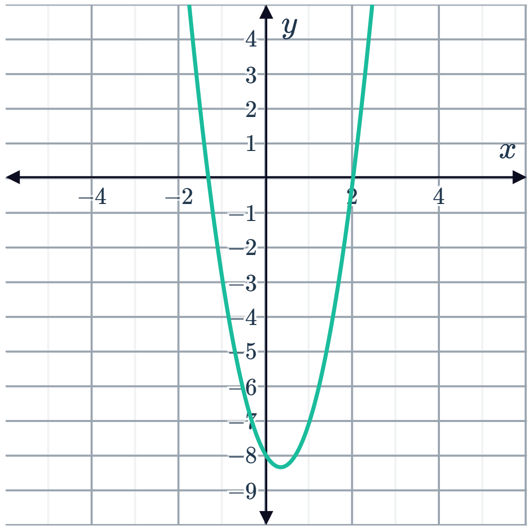


b $-1 \leq x \leq 5$

14

a $x = -\dfrac{4}{3}, 2$

b



c $x \leq -\dfrac{4}{3}$ or $x \geq 2$

15

a This parabola will have one x-intercept.

b $x = 3$

16

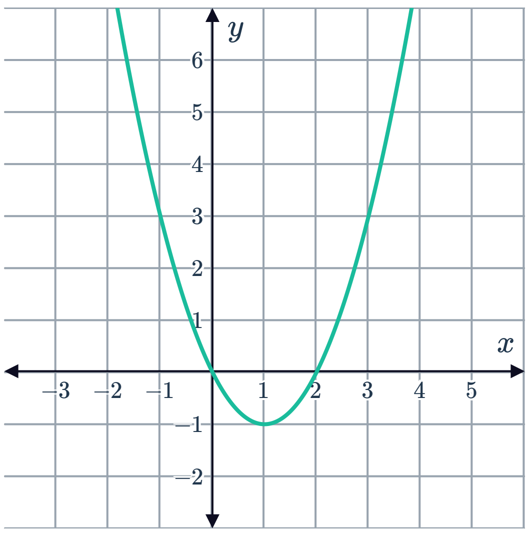
a $x = \dfrac{-9 \pm \sqrt{17}}{4}$

b Concave up

c $x < \dfrac{-9 - \sqrt{17}}{4}$ or $x > \dfrac{-9 + \sqrt{17}}{4}$

17

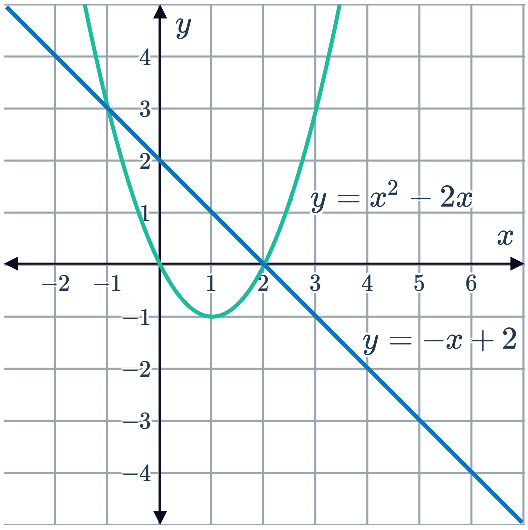
a



b $-2 \leq x \leq 4$

18

a

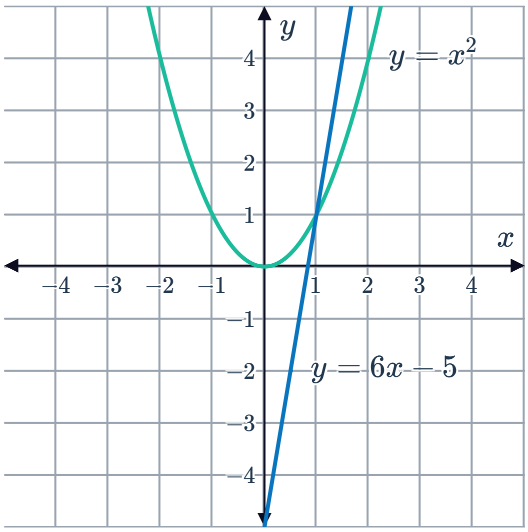


b $x = -1, 2$

c $-1 \leq x, x \leq 2$

19

a

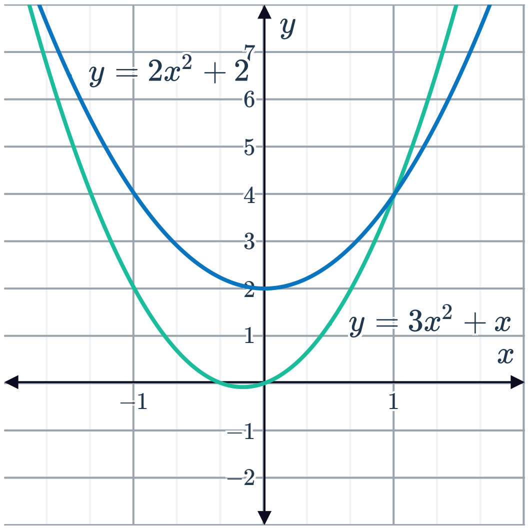


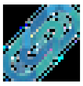
b $x = 1, 5$

c $x < 1, x > 5$

20

a





b $x \leq -2$ or $x \geq 1$

21

a $x < -3, x > 3$

b $0 \leq x \leq 2$

c $x \leq 2, x \geq 3$

d $x < -4, x > -3$

Equivalent inequalities

22 $y = 2x$

23 $y = 4x + 4$

24 $(-\infty, -3)$

25

a $x > 5$

b $[5, \infty)$

26 $(-\infty, -2]$

27

a Yes, because this inequality can be written as $f(x) \leq g(x)$.

b No, because the inequality cannot be easily written in terms of $f(x)$ and $g(x)$.

c No, because the inequality cannot be easily written in terms of $f(x)$ and $g(x)$.

d No, because the inequality cannot be easily written in terms of $f(x)$ and $g(x)$.

e Yes, because the inequality can be written as $f(x) > g(x)$.